

Workshop: Prompt Engineering in Data Analytics

Support by grant “AI Adoption in Data Analytics”

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Agenda

01

Introduction to Prompt Engineering in Data Analytics

02

Basic Prompt Elements

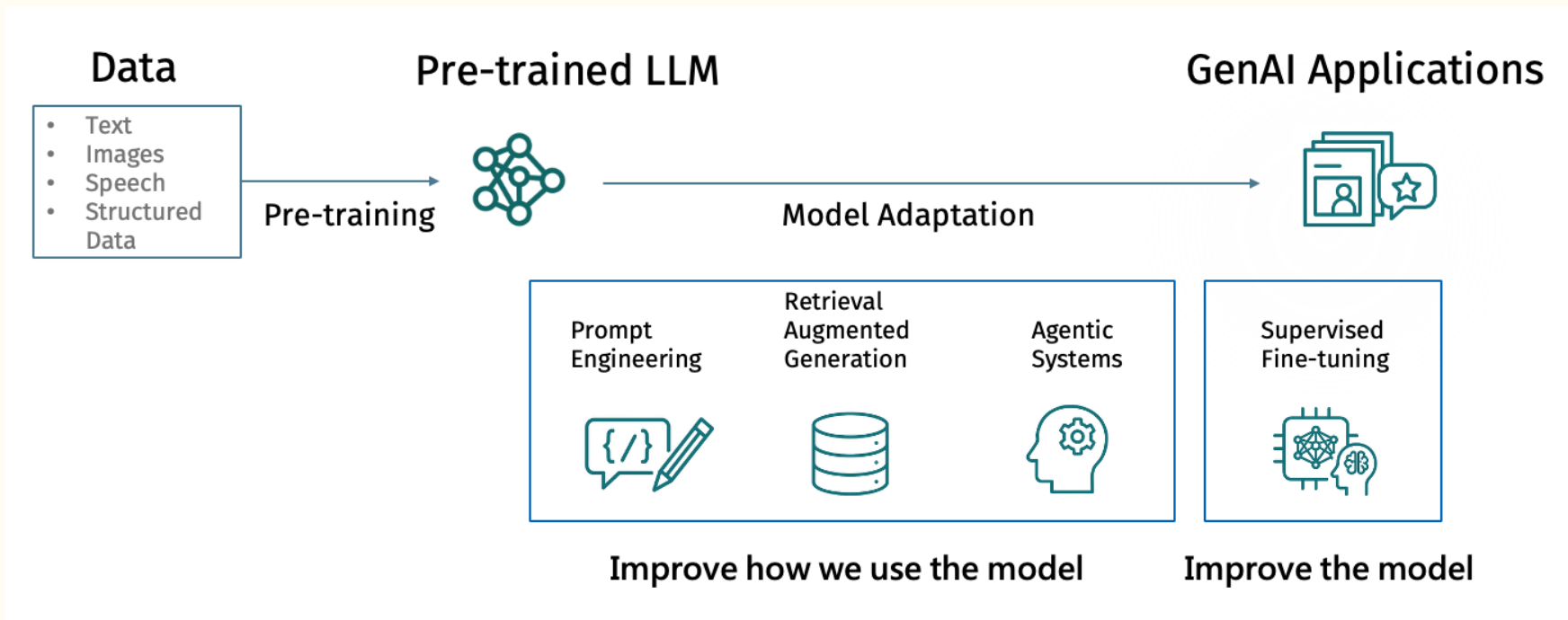
03

Common Prompt Engineering Techniques

Overview of Prompt Engineering

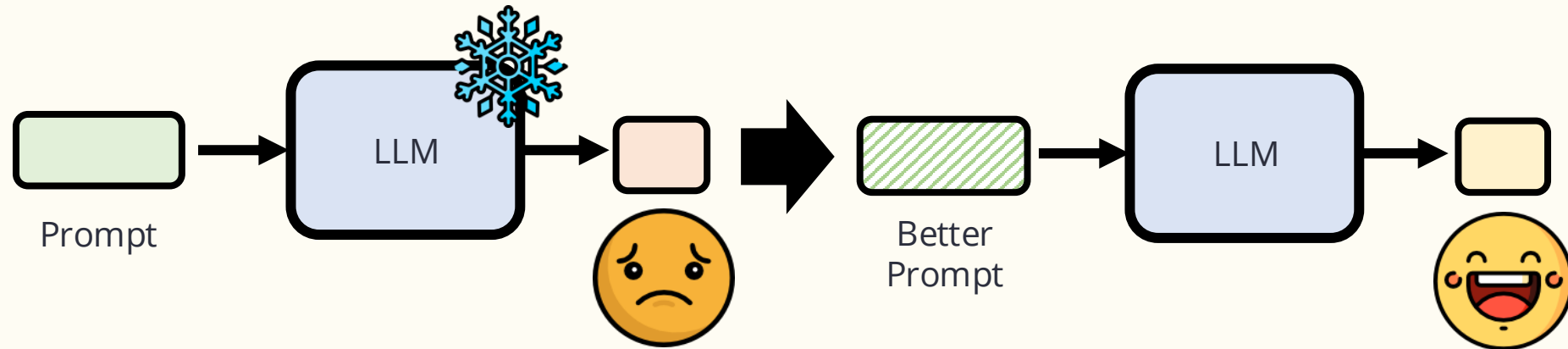
- **Key Concepts:**

- **Prompt engineering:** the skill of crafting clear text inputs (prompts) to guide GenAI models such as LLMs toward specific outputs (Brown et al. 2020).
- The art of crafting inputs to get *better outputs*.



What Is Prompt Engineering?

The art and science of humans effectively communicating with AI systems to achieve desired outcomes

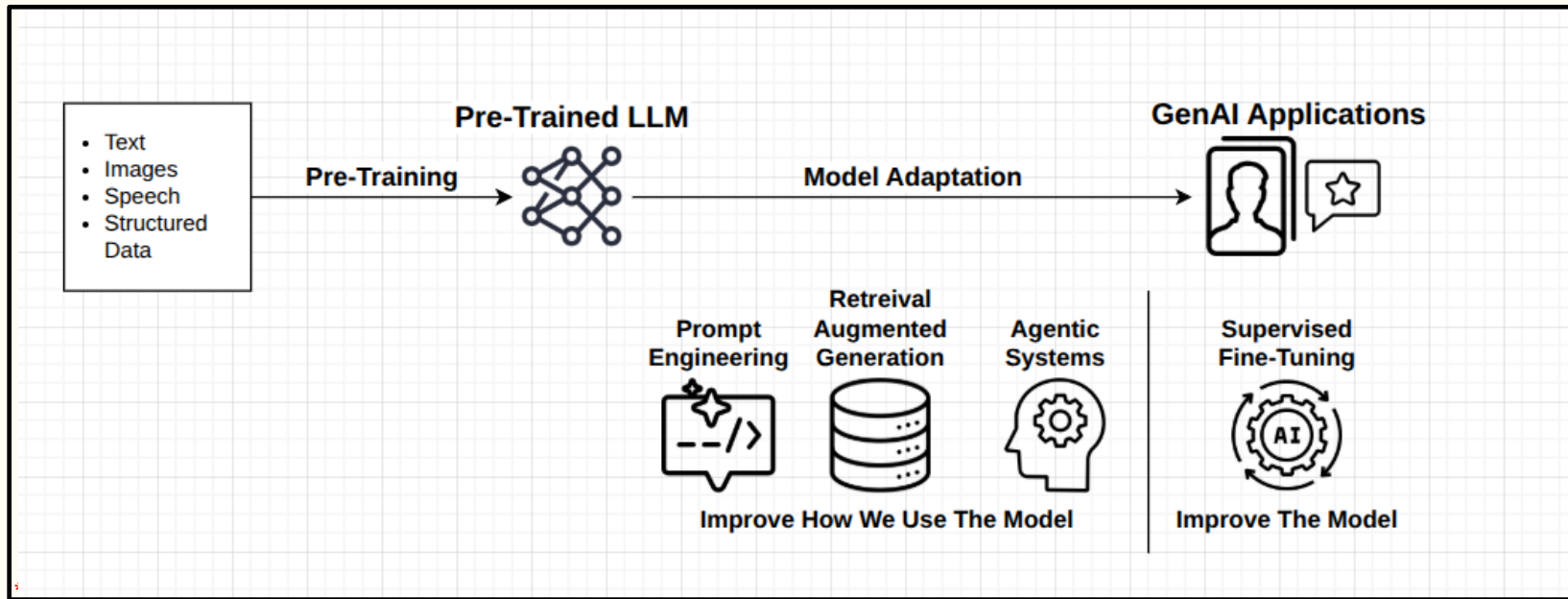


📖 **Etymology:** The term *engineering* derives from Latin *ingenium*, meaning "brilliant idea" or "flash of genius"

Core Elements of Zero-shot Prompts

Four essential elements* ensure model knows:

- ***Specific Instruction:** what to do*
- ***Context:** what background or scenario to consider*
- ***Input:** what information to process*
- ***Output:** how to format or structure the response*



A Business Application: *sentiment analysis of customer reviews*

- Imagine that you are analyzing customer reviews for a sports shoe retailer.
(Remember: *Specific Instruction, Context, Input, Output*)
- A well-crafted prompt might be:

Prompt:

Analyze the following customer reviews for our sports shoe products, focusing on feedback related to fit, material quality, durability, and style.

Summarize the three most positive aspects and the three most negative aspects mentioned in the reviews.

Include the overall sentiment, represented as the percentages of positive and negative reviews.

Conclude with a one-sentence actionable suggestion to improve customer satisfaction.

Application in Data Analytics:

Prompts for “Concept Learning”

- Suppose you want to use Gen AI to help you understand fundamental concepts.
- A possible prompt:

Prompt:

What is data mining?

Prompt:

Explain the random forest model in a simple and easy-to-understand manner.

Prompt:

What is the difference between supervised and unsupervised data mining methods?

Prompt:

Why should cross-validation be used when developing data mining models?

Application in Data Analytics:

Prompts for “Concept Learning”

- Suppose you want to develop a chatbot to help you understand fundamental concepts.
(Remember: *Specific Instruction, Context, Input, Output*)
- A possible prompt:

Prompt:

Explain the term asked in the question in 1–5 concise sentences, ensuring clarity and accuracy, using an intuitive and easy-to-understand way with examples.

Output Format:

- Short paragraph around 300 words
- Provide business applications, examples, or bullet points.
- Briefly explain why the concepts are important in industry.

Important Reminder:

- Focus on a clear, authoritative explanation suitable for a general audience.
- Begin by precisely describing the concept before giving a concise summary or definition.
- Remember: Give the detailed reasoning or description first, then the short definition or conclusion.

Comparison of Prompts

Concept-learning ▾ v2 · default ▾		⋮ @ Evaluate	Update
A	gpt-4.1 ↗	✎ ✕	B gpt-4.1 ↗ ✎ ✕
System message		System message	
Explain the concept asked		Explain the term asked in the question in a few paragraphs, ensuring clarity and accuracy, using	
User		User	
what is data mining		what is data mining	
Assistant		Assistant	
Data mining is the process of automatically discovering useful patterns, trends, correlations, or information from large sets of data using techniques from statistics, machine learning, and database systems. The main goal is to extract meaningful, actionable insights from raw data that can be used for decision-making or prediction.		Data mining is the process of discovering patterns, correlations, trends, or useful information from large sets of data, using statistical, mathematical, and computational techniques. Imagine a vast database with millions of customer transactions at a supermarket—individually, each transaction means little, but collectively, they hide valuable insights. For example, by analyzing this data, you might find that customers who buy diapers are also likely to buy wipes, which is a pattern not obvious at first glance. Data mining uses algorithms to sift through complex data, looking for these hidden relationships and making sense of raw data, which would otherwise be overwhelming due to its sheer size and complexity.	
Key Points:		Businesses use data mining for various applications, such as:	
• Large Data Sets: Works with "big data" or large databases. • Finding Patterns: Looks for relationships, trends, groups, and anomalies. • Techniques: Uses algorithms for classification, clustering, association rule mining, regression, etc. • Applications: Used in business (market analysis, fraud detection), healthcare (diagnosis, drug discovery), science, and more.		• Market basket analysis: Understanding product buying patterns. • Customer segmentation: Grouping customers by purchasing behaviors for targeted marketing. • Fraud detection: Identifying unusual transactions that may indicate fraud. • Predictive analytics: Anticipating customer churn or sales trends.	
Example:		In summary, data mining means extracting meaningful, actionable insights from large datasets using computational techniques. This is critical for companies because it helps in making informed decisions, optimizing marketing strategies, improving customer satisfaction, and gaining a competitive edge in the market by leveraging data-driven strategies.	
An online retailer can use data mining to analyze purchase histories and customer behavior to recommend personalized products or detect fraudulent transactions.			
In summary:			
Data mining turns massive amounts of raw data into useful knowledge and insights, supporting better decisions and predictions.			
👍 Good 👎 Bad		👍 Good 👎 Bad	

Application in Data Analytics:

Prompts for “Coding assistant”

- Suppose you want to use Gen AI to help you write codes.
- A possible prompt:

Prompt:

How to draw a pie chart in R for the MD variable in “aircraft” dataset?

Prompt:

How to draw a pie chart in python for the MD variable in “aircraft” dataset?

Prompt:

How to explore the relationship between a car’s weight and its fuel efficiency in the “mtcars” dataset?

Application in Data Analytics:

Prompts for “Coding assistant”

- Suppose you want to have an assistant to solve some coding tasks.
- (Remember: *Specific Instruction, Context, Input, Output*)
- A possible prompt:

Prompt:

In the "flights" dataset from the "nycflights13" package, write the codes to filter all the flights that are

1. in January, July, and December,
2. the destination is "BWI",
3. the arrival delay is greater than or equal to 60 minutes.

Application in Data Analytics:

Prompts for “Coding assistant”

- Try the other questions in your lab 2

Your Prompt:

Application in Data Analytics:

Prompts for “Practice Tests”

- Sample prompts:

Prompt:

Can you please give me some practice questions on basic R language syntax?

Prompt:

give me a practice question on data wrangling using dplyr package? For example, filtering observations

Prompt:

please give me some questions about neural network models?

Application in Data Analytics:

Prompts for “Practice Tests”

- Think about: *Specific Instruction, Context, Input, Output*

Prompt:

Help me study some basic concepts *in [data mining and business analytics]*.

Write a sentence for me to fill in the blank, and I'll try to pick the correct option for the *[data mining or business analytics]* concepts.

Check my answers and correct me if I'm wrong and explain why.